

Antonio D'Alessandro

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Education

Ph.D. in Biostatistics, New York University Expected 2026

Dissertation: *Modern Causal Machine Learning for Data Integration and Treatment Effect Heterogeneity*

Advisors: Michele Santacatterina and Samrachana Adhikari

M.S. in Biostatistics, New York University 2025

B.A. in Mathematics, Queens College 2022

Research Interests

My research focuses on the development and application of causal inference and data science methods using both experimental and observational data. Past projects have included methods for improving precision when estimating heterogeneous treatment effects, developing new doubly robust methods for time-to-event outcomes in platform designs, and new methods for longitudinal studies with missing data. My research interests include:

- causal inference
- machine learning
- semiparametric theory
- bayesian statistics
- sports analytics

Industry Experience

Data Scientist 2019–2022

Manhattan Beer Distributors, Bronx, NY

- Developed predictive models for operational forecasting and resource allocation to support business decision-making.
- Built automated analytics pipelines in R and Python, significantly reducing manual reporting effort.
- Designed and evaluated A/B tests and observational studies for evaluation of operational interventions.
- Led cross-functional analytics initiatives with engineering and operations stakeholders.

Data Analyst 2012–2019

Manhattan Beer Distributors, Bronx, NY

- Built statistical models and predictive systems using R, Python, and SQL.
- Developed reproducible data pipelines and version-controlled workflows using Git.
- Conducted exploratory analysis, model validation, and performance evaluation for decision support.
- Communicated insights via dashboards, reports, and technical presentations.

Research Experience

Graduate Researcher, Department of Biostatistics 2022–Present

New York University, New York, NY

- Develop causal machine learning methods for treatment effect heterogeneity in randomized controlled trials.
- Investigate semiparametric and doubly robust estimators for subgroup-specific treatment effects.
- Develop methodology for causal survival analysis in platform trials with non-concurrent control arms.

- Study data integration methods that combine randomized and observational data sources.
- Conduct large-scale simulation studies evaluating finite-sample performance of causal estimators.
- Implement statistical methods in R and Python using modern machine learning and semiparametric estimation techniques.
- Collaborate with faculty researchers on methodological and applied statistics projects.

Selected Publications

D'Alessandro, A. et al., Modern Causal Inference Approaches To Improve Power For Subgroup Analysis In Randomized Controlled Trials. *Statistics in Medicine*. <https://arxiv.org/abs/2505.08960>

D'Alessandro, A., Adhikari, S., Santacatterina, M., Causal Survival Analysis in Platform Trials with Non-Concurrent Controls (Under review). <https://arxiv.org/pdf/2603.10989>

Susmann, H., D'Alessandro, A., The Counterfactual Combine: A Causal Framework for Player Evaluation (Under review). <https://arxiv.org/pdf/2602.23233>

Teaching Experience

Teaching Assistant Fall 2024

Introduction to Probability and Statistics

New York University

Teaching Assistant Spring 2021

Introduction to Bayesian Statistics

Queens College

Technical Skills

Programming Languages

R, Python, SQL

Statistical Methods

Modern Causal Inference, Machine Learning, Survival Analysis, A/B Testing, Bayesian Statistics, Semiparametric Estimation, Doubly Robust Estimation

Software and Libraries

R, tidyverse, ggplot2, tidymodels, SuperLearner, ranger, gbm, xgboost, BART, survival, lme4, future, Stan, Python, NumPy, pandas, SciPy, scikit-learn, PyMC, PyTorch, TensorFlow, Jupyter, Git, GitHub, Quarto, R Markdown, MLOps, Agentic Workflows, AWS, Snowflake, Shiny, Tableau, Google Cloud